Diagnosis-Related Groups (DRG)

Shadow Billing Phase

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هيئة التأمين
Insurance Authority

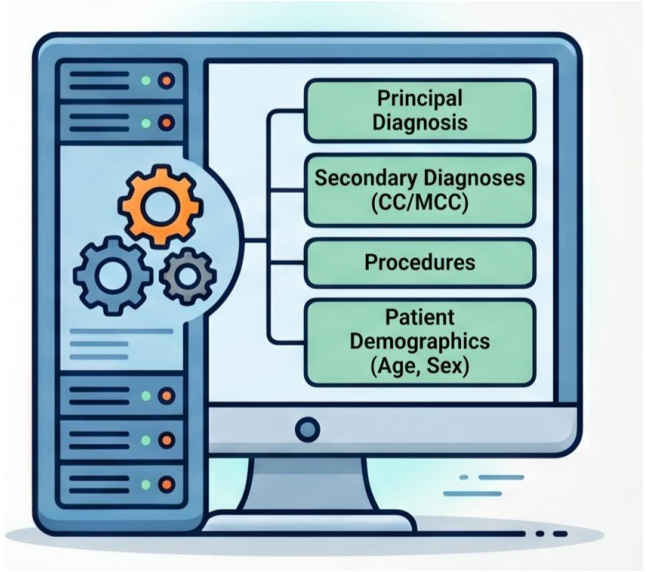
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What is Diagnosis Related Groups

A clinical classification system that groups inpatient and day-case patients into packages (groups) based on the similarity in terms of clinical complexity and the resources used to treat it. Most DRG has grades of complexity splits from the most complex (A split) to the least complex (D Split)



DRGs are assigned by the medical coders using a software called Grouper based on the information collected from the medical record



The main drive for DRGs is Diagnoses (Principal and Secondary) but it may be affected surgical and non-surgical procedures and some demographic data such as age and gender, and administrative data such as LOS and discharge status

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Fee for Service Reimburesement

The more services provided, the higher the bill



Total Services Provided

Doctor's examination, inpatient days, medical examinations, tests, radiology, surgery....

=



Case Price

The total amount paid by the payer

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DRG Reimbursement Equation (Case Price = Relative Weight x Base Price)



Case Price

The total amount paid by the payer



Relative Weight

A digital indicator that reflects the severity of the condition and the resources used to treat it



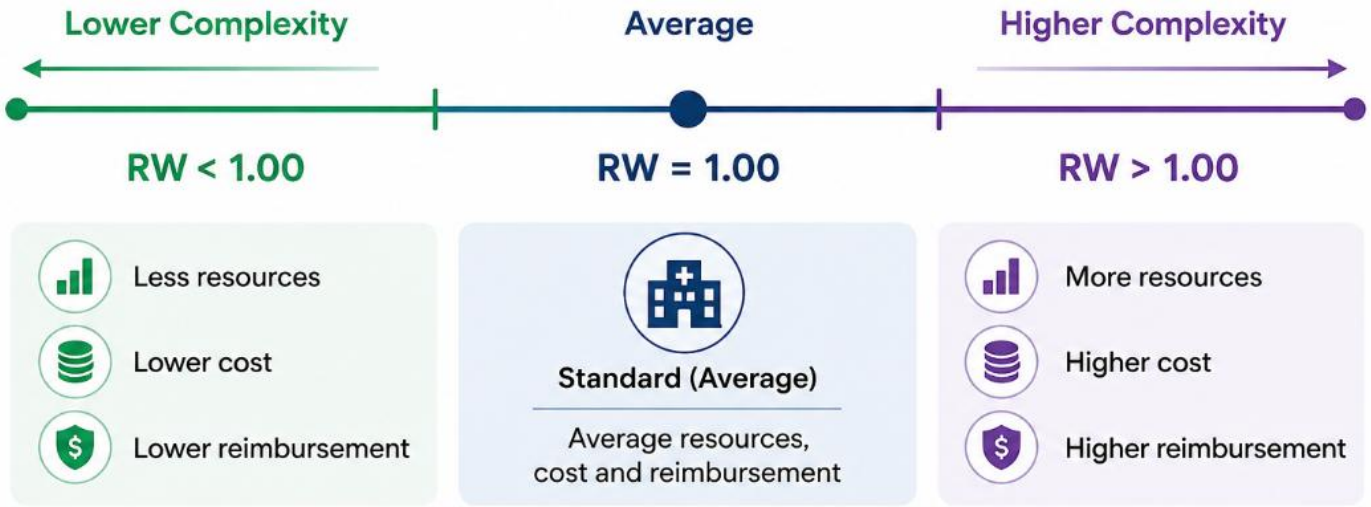
Base Rate

Fixed financial value reflecting the average cost of hospital treatment

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Case Price = Relative Weight × Base Rate

Relative Weight (RW): A value assigned to each DRG that reflects the average resources and cost required to treat a patient relative to the overall average.





Relative Weight

Relative Weight (RW) Reflects

- **Resource Consumption:** Staff time, medications, diagnostics, procedures, and length of stay.
- **Hospital Reimbursement:** Payment = Base Rate × RW
- **Clinical Complexity:** More severe cases generally have higher RWs.
- **Case Mix Index (CMI):** Average of all RWs is used to assess hospital complexity and support benchmarking.

Higher RW ⟶ Higher Resource Use ⟶ Higher Payment

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Case Price = Relative Weight × Base Rate

Base Rate: A fixed monetary value representing the average cost of treating a patient with average complexity (RW = 1.00).

It is a key factor in calculating the amounts insurers or government bodies pay the hospital

Countries may adopt either a single national base rate or a range of base rates, depending on their payment model.

Factors that influence the base rate:

- Case Mix Index (CMI)
- Hospital type (e.g., teaching/tertiary)
- Geographic location & operating costs



Base Rate

Fixed financial value reflecting the average cost of hospital treatment

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DRG Reimbursement – other factors

Case Price = Relative Weight × Base Rate × (Adjustment Factors + Other Add-ons)

Adjustment Factors & Add-ons: These factors aim to protect hospitals from financial losses in highly complex cases, encourage quality of care, and prevent gaming of the system.

- **Length-of-stay outliers:** when a patient stays longer or shorter than the average length of stay defined for the DRG.
- **Quality incentives:** factors added to reward providers for adopting efficiency standards.
- **Quality penalties:** deductions for low quality, such as readmissions and hospital-acquired conditions and complications like infections and injuries.
- **Cost outliers:** when a patient consumes very expensive resources and medications, or items whose prices vary widely and are hard to capture in the DRG relative weight, including new technologies such as robotics.



Case Price

The total amount paid by the payer

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How the Diagnosis-Related Groups system works

Comparison Between the Two Payment Systems

	Fee-for-Service (FFS)	Diagnosis-Related Groups (DRG)
Payment model	Payment varies regardless of the complexity of the case	A fixed amount per DRG regardless of the services provided
Unit of measurement	Services Provided	DRG code
Cost predictability for the payer	Unpredictable	Predictable based on the diagnoses of hospitalized patients
Cost predictability for the provider	Unpredictable and dependent on approvals	Predictable based on medical case criteria
Efficiency incentive	None	Reduction in length of stay and unnecessary services
Risk of waste and misuse	Very high	None – No additional payments for performing unnecessary services.
Financial Risk Bearing	High risk for insurance companies	Lower risk, shared between providers and payers, with a defined management strategy.
Quality of services	Balanced	There is a risk of reduced quality; therefore, service quality must be monitored (e.g., length of stay and readmission rates)

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Why implementing the Diagnostic-Related Groups (DRG) system improves medical quality and efficiency ?

	Payment per Volume of Services (Fee-for-Service)	Payment per Diagnostic Related Groups (DRG)
Incentives	More services and longer hospital stays	Increased efficiency
Length of hospital stay	Longer	Shorter by 0.8 to 1.5 days
Hospital readmissions	No Impact	Lower incentive
Cost per visit/admission	Increases with the number of services and days	Lower or must be controlled
Negotiations & Contracting	Complicated & Lengthy Long list of services & packages	Easier & Shorter Base Rates
System efficiency	Weak	Improvement ranging from 5% - 10% ↑

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How AR-DRGs Determine Hospital Payment

Base rate for the healthcare provider (negotiated) = 10,000 SAR

(Payment for an average case)

Example: Two patients admitted for the same condition with different levels of complexity

More Complex Case

AR-DRG: E62A

Respiratory infections and inflammatory conditions

Diagnoses:

Primary diagnosis: Pneumonia
Secondary diagnoses :
Chronic hypertension , Hypothyroidism
Procedures:

- Respiratory imaging

Relative Weight: 1.7311

Payment = Relative Weight × Base Rate

1.7311 × 10,000 = **17,311 SAR**

Less Complex Case

AR-DRG: E62B

Respiratory infections and inflammatory conditions

Diagnoses:

Primary diagnosis: Pneumonia
Secondary diagnoses :
None
Procedures:

- Respiratory imaging

Relative Weight: 0.7599

Payment = Relative Weight × Base Rate

0.7599 × 10,000 = **7599 SAR**

As case complexity increases → The relative weight increases → Higher payment amounts

***Relative Weights are set and mandated by the regulator**

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How AR-DRGs Determine Hospital Payment

Base rate for the healthcare provider (negotiated) = 10,000 SAR

(Payment for an average case)

Example: Two Patients admitted for different conditions:

More Complex Case

AR-DRG: B70B

🧠 Stroke & Other Cerebrovascular Disorders

Procedures:

- Thrombolysis (tPA administration)
- CT/MRI Brain Imaging
- Neurological Monitoring
- Rehabilitation Services

Relative Weight: 1.99

Payment = Relative Weight × Base Rate

1.99 × 10,000 = **19,900 Riyal**

Less Complex Case

AR-DRG: C16Z

👁 Lens Procedures

Procedures:

- Cataract extraction (phacoemulsification)
- Intraocular lens implantation
- Pre-operative assessment
- Day surgery care

Relative Weight: 0.82

Payment = Relative Weight × Base Rate

0.82 × 10,000 = **8,200 Riyal**

As case complexity increases → The relative weight increases → Higher payment amounts

***Relative Weights are set and mandated by the regulator**

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How AR-DRGs Determine Hospital Payment

The base price may vary for the same service provider in contracts with insurance companies based on negotiations

Example: Reimbursement for the same case (fixed relative weight) from different insurance companies

Health Insurance Company 1

Base Price = 10,000

AR-DRG: B70B

🧠 Stroke & Other Cerebrovascular Disorders

Relative Weight: 1.99

Payment = Relative Weight × Base Rate

1.99 × 10,000 = **19,900 Riyal**

Health Insurance Company 2

Base Price = 12,000

AR-DRG: B70B

🧠 Stroke & Other Cerebrovascular Disorders

Relative Weight: 1.99

Payment = Relative Weight × Base Rate

1.99 × 12,000 = **23,880 Riyal**

Reimburesement for the same case may vary depending on the contract with insurance companies or the government payer (Center for National Health Insurance).

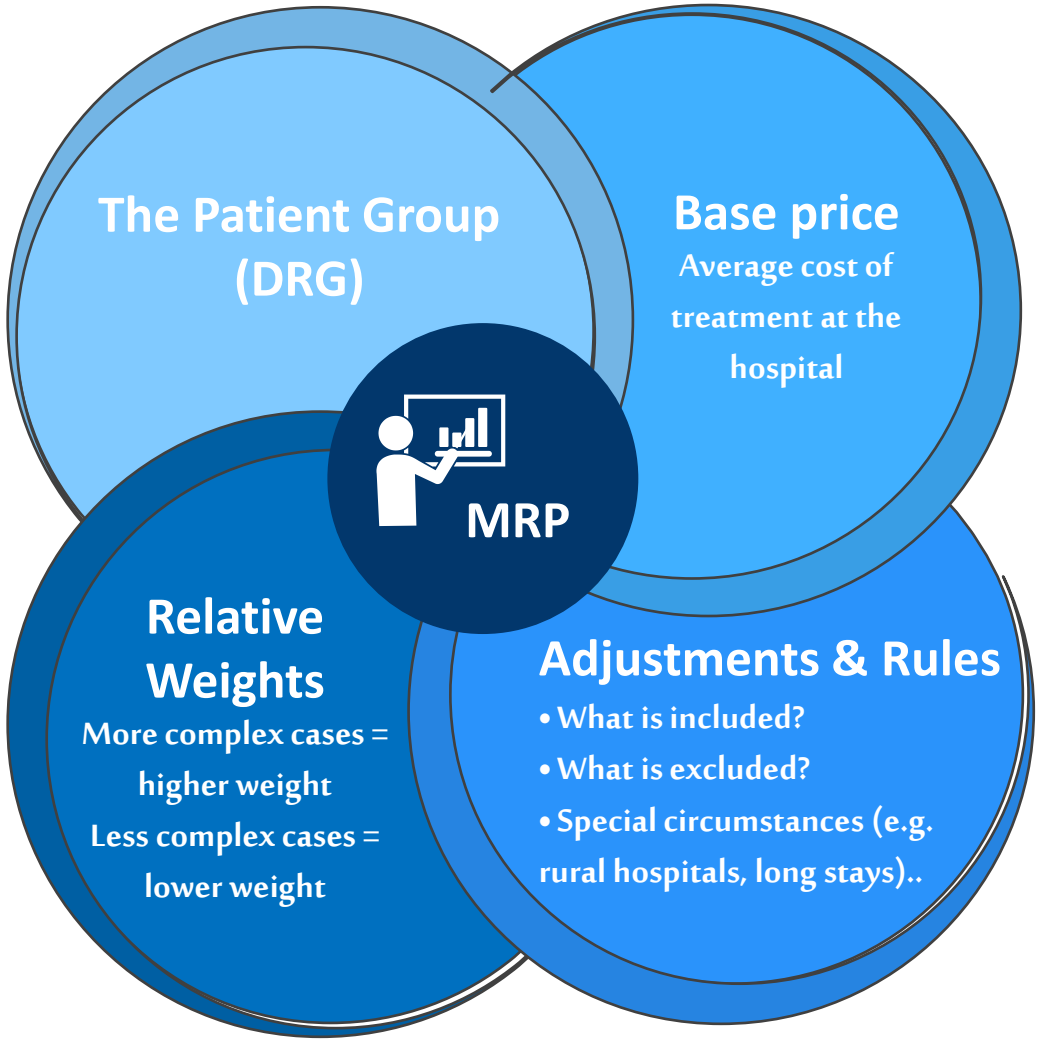
*Relative Weights are set and mandated by the regulator

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Market Reference Price (MRP)

The Market Reference Price (MRP) A **non-mandatory** indicative benchmark price range (minimum–median–maximum) for health services, that serves as a reference to promote market competitiveness and transparency.

Data source:
The Analysis was based on claims submitted by providers through the NPHIES system during the period from January 2022 to May 2023, considered as an indicative proxy for cost estimation



- **Periodical updates to reflect inflation rates and changes in the market**
- **Annual reviews for the first 3-years and then the review period is re-evaluated**

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Shadow Billing Objectives

Shadow billing involves using both the existing fee for service pricing system and the new AR-DRG reimbursement system simultaneously. This is a testing phase where selected payers and providers will apply AR-DRG to better understand case mix of services provided and potential financial implications prior to full implementation.

Parallel collection claims

- Fee for Service
- AR-DRG Reimbursement

This approach helps to collect accurate data, report on any financial changes, and model the impact of changes that may occur when implementing AR-DRG system.

Assess readiness of the health system
Data, systems, people and processes

Assess financial impact

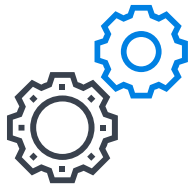
Monitor changes in the system
For example, behaviour changes by provider

Facilitate development and prep for full implementation

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Why participate in Shadow billing

Shadow billing is a critical step before full transition allowing Providers to learn, prepare and refine before final AR-DRG transition



Enables preparation

Allows providers to understand requirement for DRG reimbursement and allows for preparation for full implementation



Training of staff

An opportunity for providers and payers to train their staff through participation



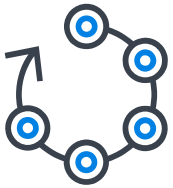
Financial Impact Assessment

Results from Shadow billing will be useful to understand the impact of the changes in DRG reimbursement



Transparent Engagement

Providers will be able to engage with CHI and payers to contribute in transparent discussions to refine and shape the final DRG reimbursement mechanism



Process Refinement

Providers will be able to review and refine their processes before actual transition into DRG reimbursement



Systems updates

Enable providers to update systems for integration with DRG groupers, updated claim submissions in line MDS.



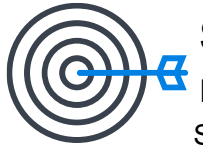
Time to adjust to changes

Shadow billing is a transition phase, allowing time to learn, implement changes and adapt to new reimbursement mechanism



Facilitates Negotiation

Shadow billing is a transition phase, allowing time to learn, implement changes and adapt to new reimbursement mechanism



Strategic decision making

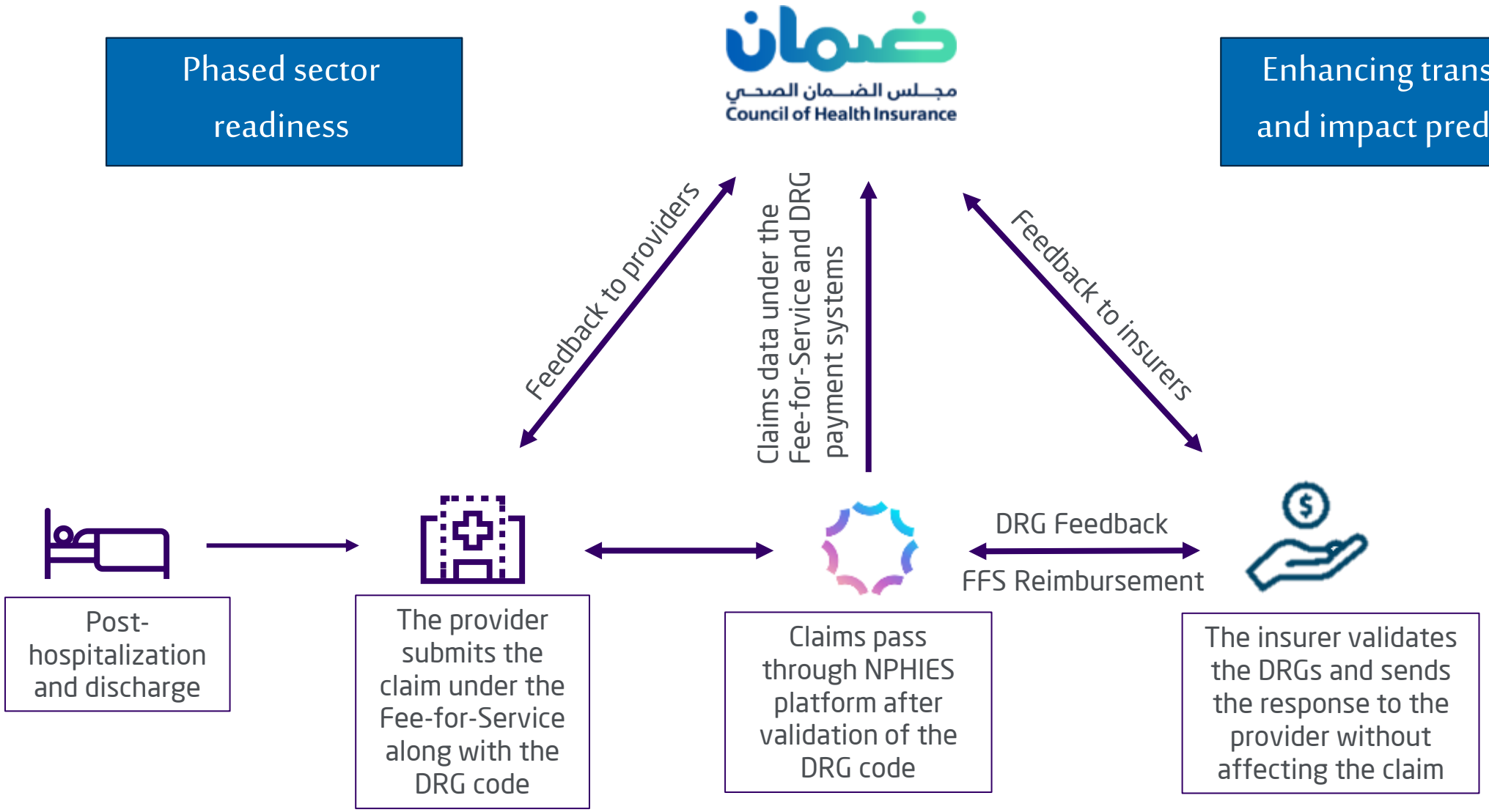
Shadow billing is a transition phase, allowing time to learn, implement changes and adapt to new reimbursement mechanism

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The Shadow Billing Process flow

Phased sector readiness

Enhancing transparency and impact predictability



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Shadow Billing Implementation Timeline

The implementation roadmap for DRG Shadow Billing, outlining the key milestones leading to production deployment within NPHIES.



Workshop

Present the Shadow Billing program, including its objectives, implementation approach, and roadmap

Align providers and payers on participation and next steps.

5 July 2026



Guidelines

Publish the DRG guideline and supporting implementation materials.

Support providers and payers in preparing for Shadow Billing implementation.

5 July 2026



Technical Workshop

Technical workshop on the updates required in the integration and provider systems.

Walk through DRG code submission within **NPHIES**.

12 July 2026



Testing

Begin end-to-end testing in the NPHIES testing environment.

Validate DRG submissions and resolve identified issues before production.

13-31 July 2026



Go-Live

Begin DRG Shadow Billing submissions in the NPHIES production environment.

Support the transition to live Shadow Billing operations.

2026 - TBD

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DRG Implementation Plan

Implementing Diagnosis-Related Groups (DRG) system requires a phased approach, beginning with sector preparation and readiness, followed by testing, Shadow Billing, implementation readiness, and finally full implementation.



Pilot Phase

First Half

- Data completeness and NPHIES readiness
- Coordination with payers (Insurance Authority and National Center for Health Insurance)
- Announce the shadow-billing start date
- Launch sector awareness and training workshops

Second Half

- **Begin the shadow-billing phase (Q3-Q4)**
- Continue fee-for-service payment
- Launch the Market Reference Price (MRP) and payment calculator
- Sector feedback
- Monitor data quality and compliance
- Sector awareness and training (ongoing)

Implementation Readiness

- Continue shadow billing
- Update the Market Reference Price
- Assess readiness for full implementation
- Support DRG-based contracting

Full Implementation

- Begin reimbursement using DRGs
- Monitor and measure data quality
- Update the market reference price (MRP) annually

The market reference price (MRP) is non-binding and subject to negotiation**

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DRG Implementation Guidelines

Practical guidance to support DRG readiness and implementation across the private health sector

- ✓ Shadow Billing requirements & process
- ✓ Clinical documentation & coding
- ✓ DRG grouping, billing & validation
- ✓ MRP, reporting & contracting
- ✓ Readiness criteria & checklist



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اللوائح والسياسات

الأنظمة واللوائح
السياسات
معايير التصنيف الطبي
القرارات والتعاميم
وثائق الضمان الصحي
المبررات الطبية
دليل ضمان للأدوية

معايير التصنيف الطبي | الموردين وشركات البرامج | التعليم والتدريب | الدعم

هذه الصفحة تحتوي على مواد متعلقة بـ:

- تطوير وصيانة معايير التصنيف للفوترة الطبية والترميز الطبي.
- تقديم الدعم والإرشاد للسوق بما في ذلك مقدمو الرعاية الصحية ، شركات التأمين الصحي، شركات إدارة مطالبات التأمين, شركات إدارة دورة الإيرادات, ومزودو نظم إدارة المعلومات الصحية.

نظام الحزم التشخيصية | نظام الفوترة السعودي | تدقيق الترميز الطبي | تحسين جودة التوثيق السريري

ما هو نظام الحزم التشخيصية؟

يدعم نظام الحزم التشخيصية الرعاية الصحية المبنية على الكفاءة والجودة من خلال منح القيمة لقطاع التأمين الصحي السعودي لتحقيق إدارة أفضل للمستفيدين، حيث يكمن الهدف في دعم الرعاية الصحية المبنية على الكفاءة والجودة من خلال تحقيق التميز في الحالات المجمعة وتطبيق نظام الحزم التشخيصية وتحسين الشفافية وتمكين الإبداع وتعزيز الكفاءة في القطاع من خلال ما يلي:



- بناء الوعي والمعرفة.
- تحسين جودة ترميز نظام التصنيف الدولي للأمراض والمشاكل الصحية ذات الصلة.
- إعداد القطاع لتطبيق نظام الحزم التشخيصية.
- تقييم تأثير تنفيذ نظام الحزم التشخيصية.

الرسالة

- هدفنا هو تحقيق التميز في مزيج الحالات وتطبيق الحزم التشخيصية.

الفوترة التجريبية باستخدام مجموعات الحزم التشخيصية اضغط هنا

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إرشادات تطبيق نظام الدفع بالحزم التشخيصية

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Supportive Tool – DRG Service on SEHA Platform

New Supportive Service

A DRG web-based service to support the sector

Lean Business Services is introducing a DRG service, a supportive tool for providers and payers delivered through the **SEHA platform** to help the sector adopt and operate DRG with confidence

Enter the **encounter details** and the service instantly **calculates the DRG** and provides the result.

An official announcement will be published on the **SEHA platform** once the service is released.

Release Date

End of July 2026

The DRG service goes live on the SEHA platform, available to the sector as a supportive tool.

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Q&A

Thank You

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